

Mining Rich Data Types

- ☐ Mining Text Data 
- ☐ Mining Spatial-Temporal Data
- ☐ Mining Graph and Networks

Mining Text Data

- ❑ Over 80% of the data is in an unstructured format, e.g., news, image, audio.
- ❑ Goal: deriving high-quality information, such as structures, patterns, and summaries, from **unstructured** text data.
 - ❑ Process: mining structures from text $\Rightarrow$ deriving patterns within the structured data $\Rightarrow$ evaluating and interpreting output
- ❑ Main challenge:
 - ❑ How to represent text primitives (e.g., words, sentences, and documents) and how to learn effective representation from unstructured text?

Mining Text Data: Major Tasks

- ❑ Information extraction (IE): automatically extracting structured information from unstructured and semi-structured documents.
- ❑ Taxonomy construction: constructing different multi-faceted taxonomies based on different corpora or applications.
- ❑ Text clustering: grouping unlabeled texts with similar contextual patterns.
- ❑ Text classification: analyzing open-ended texts (e.g., webpages) and assigning them a set of predefined topics or categories
- ❑ Text summarization: creating a subset of text that represents the most important information from a single or multiple pieces of text.

Mining Text Data: Techniques

- ❑ Symbol-based representation: using one-hot vectors for words and bag-of-words model for documents.
 - ❑ Issues: high-dimensionality and data sparsity
- ❑ Distributed representation: low-dimensional real-valued dense vectors with large amount of unlabeled data using deep neural network.
 - ❑ Word embedding: words similar in meaning are expected to be closer in the embedded space.
 - ❑ Pre-trained language model: using large-scale data and more computing resources to generate contextualized embeddings.

Mining Text Data: Techniques

- Word2vec: words that share common contexts are embedded in close proximity, e.g., "Sweden" can be embedded close to Norway, Finland.
 - CBOw: predicts the current word from surrounding context words.
 - Skip-gram: uses the current word to predict the surrounding words.
- GloVe: combining global matrix factorization and local context window methods.
- JoSE: learning word embeddings in hyperbolic space to model hierarchical structures.
- Pre-trained language model: ELMo and BERT

